

List of student internships offered by newcleo

–academic year–
2025–2026

February 18, 2026

Preface

This document contains the internship offers supported by *newcleo*. Each offer contains the specifications and modalities for the application and execution of the internship work.

Please submit your CV and motivation letter to the contact person indicated in the internship offer.

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About newcleo

Privately funded and headquartered in London, *newcleo* was launched in 2021 – and since raised a total of EUR 400m – to be an innovator in the field of nuclear energy. Its mission is to generate safe, clean, economic and practically inexhaustible energy for the world, through a radically innovative combination of existing, accessible technologies.

With visionary co-founders, *newcleo* capitalises on thirty years of R&D activity in metal-cooled fast reactors and liquid-lead cooling systems, and its senior management and advisory team can boast hundreds of years in cumulative hands-on experience.

Counting on around 800 highly skilled employees across Europe, *newcleo* has business, scientific, operations and industrial manufacturing capabilities in a vertically integrated model designed to deliver its ambitious timeline for its plan-to-market.

newcleo's technology, mostly comprising a novel approach to already qualified solutions, addresses equally well the three challenges affecting the nuclear industry to date:

- **Waste:** fast reactors are capable of efficient “burning” (i.e., fission) of depleted uranium, plutonium and minor actinides. When operated with mixed-oxide (MOX) fuel generated from reprocessed nuclear waste, *newcleo's* reactors not only ensure sustainability by closing the fuel cycle, but can also boost energy independence;
- **Safety:** lead-cooled reactors operate at atmospheric pressure. The properties of lead (thermal capacity and conductivity, boiling point, chemically inert, low neutron activation, shielding properties) together with *newcleo's* passive safety systems ensure very high levels of safety;
- **Cost:** *newcleo's* reactor design has been optimised over the last 20 years leading to the concept of an ultra-compact and transportable 200MWe module with improvements in energy density compared to other technologies. Costs are kept low by means of simplicity, compactness, modularity, atmospheric pressure operation and elevated output temperature.

newcleo is also working to significantly invest in MOX fuel manufacturing in developed countries, extracting energy from the current nuclear industry by-products.

newcleo is ready to develop a new, sustainable, and completely safe way of generating nuclear energy that will help humanity reach zero emissions and mitigate global warming.

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Comparison of numerical acceleration methods for nonlinear iterations used in the computer code TRANSURANUS

Elena Travaglia and Daniele Tomatis*

newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

January 8, 2026

– *Internship proposal* –
(ID No. 1003986)

Keywords — TRANSURANUS, numerical acceleration, non-linear iterations

Topics — fuel performance, numerical mathematics, computational modelling

Location *newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy*

Starting date To be defined

Duration 4–6 months

Context

newcleo is currently in the process of developing advanced Lead-cooled Fast Reactor (LFR) units tailored for small and modular reactors. These units incorporate innovative features aimed at addressing the unique challenges associated with LFR technology. Simultaneously, the company is committed to meeting sustainability, economic, and safety objectives, aligning with the criteria set for the fourth generation of nuclear reactors.

In the context of fuel performance analysis, *newcleo* relies on the TRANSURANUS (TU) thermo-mechanical computer code, which is distributed by the Joint Research Center of the European Commission in Karlsruhe, Germany, under a collaboration research agreement [1, 2]. *Newcleo* has initiated a meticulous examination of the code to ensure its suitability and adaptability for applications involving LFR technology. The goal is to employ precise physical models that guarantee reliable results in numerical simulations. This ongoing effort also includes improving the code to broaden its scope and enhancing its performances.

*Contact person, daniele.tomatis@newcleo.com

Description

The thermal analysis of an integral fuel rod can be treated by superposing one-dimensional radial and axial solutions. Given the multitude of non-linearities inherent in these processes, only numerical solution techniques prove feasible. These techniques are of critical importance as they determine numerical stability and, to a considerable extent, the overall computational cost.

TU solves heat transfer by the Fourier equation in the radial coordinate of the fuel rod, whose discretized form brings to a tridiagonal system. The nonlinearity of the coefficients in the second order differential equation is handled by successive substitutions. The resulting non-linear iterations are accelerated by minimizing the residuals through a numerical scheme based on Regula Falsi, initially implemented by Lassmann [3].

This procedure was adopted because calculating the temperature of a fuel rod, which is schematically shown in Fig. 1, is a computationally intensive task. Therefore, it is important to focus on obtaining convergence and minimizing the total numerical effort required.

In the current internship the student will perform a complete literature review of the existing models to solve heat conduction equation and technique to accelerate the convergence of such numerical method. Some of this technique will be implemented and their performance will be compared.

The student will be introduced to the use of TU and will join the team of TU developers at newcleo, learning the best practices for collaborative work and quality-assured code development. The production of a technical report is expected at the end of the internship to describe the work done by the student and to compile the analysis of the results.

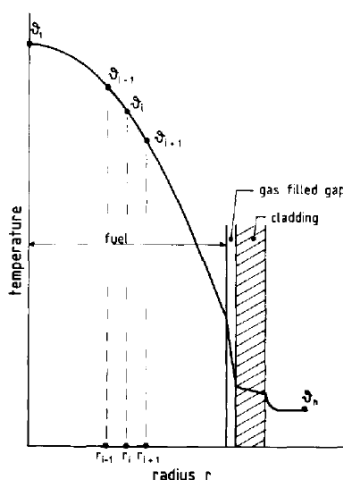


Figure 1: Schematic representation of the stationary radial temperature profile in a slice of the fuel rod

Work plan

- Literature review and retrieval of the existing work
- Discussion of the problem
- Implementation and verification of the numerical solution

- d) Analysis of the numerical results
- e) Preparation of a technical report

Applicant profile

Master or PhD student in Nuclear, industrial or Software Engineering; Applied Physics; Mathematics.

Required computer skills: Basic knowledge of Linux operative system and software programming (previous experience in FORTRAN is appreciated, but not fundamental).

Knowledge of $\LaTeX$ editing is appreciated, but not required.

References

- [1] K Lassmann. Transuranus: a fuel rod analysis code ready for use. *Journal of Nuclear Materials*, 188:295–302, 1992.
- [2] K Lassmann and H Blank. Modelling of fuel rod behaviour and recent advances of the transuranus code. *Nuclear Engineering and design*, 106(3):291–313, 1988.
- [3] K. Lassmann. A fast and simple iteration scheme for the temperature calculation in a fuel rod. *Nuclear Engineering and Design*, 103(2):211–214, 1987.

Analysis and verification of the burnup model implemented in the computer code TRANSURANUS

Elena Travaglia and Daniele Tomatis*

newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

January 8, 2026

– *Internship proposal* –
(ID No. 1003987)

Keywords — TRANSURANUS, Burnup model, LFR

Topics — nuclear fuel design, fuel performance and thermomechanics, computational modeling

Location *newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy*

Starting date To be defined

Duration 5–6 months

Context

Currently, *newcleo* is designing advanced Lead-cooled nuclear Fast Reactors (LFR). These units incorporate innovative features aimed to address the unique challenges associated with the decarbonization of the energy market and reprocessing of spent nuclear fuel. Concurrently, the company is dedicated to fulfilling sustainability, economic, and safety goals, in line with the criteria established for the fourth generation of nuclear reactors.

In the domain of fuel performance analysis, *newcleo* relies on the TRANSURANUS (TU) thermo-mechanical computer code, which is provided by the Joint Research Center of the European Commission in Karlsruhe, Germany, under a collaboration agreement [1, 2]. *newcleo* is performing a thorough examination of the code to check that it can reliably reproduce the physical behavior of LFR fuel rods.

This ongoing endeavour also includes improving the code to broaden its range of applications, with a specific focus on providing trustworthy predictions for modeling LFR fuel rods. The final goal of this initiative is to develop an improved version of the code.

*Contact person, daniele.tomatis@newcleo.com

Description

The performance of a fuel rod is greatly influenced by several physical phenomena. The exposure of the fuel rod to neutron irradiation determines the time evolution of the isotopic concentrations, and of the power density distribution within the fuel. The power density is the main driving force of the local temperature profile. The temperature heavily affects material properties and many physical phenomena, such as fuel restructuring, actinide redistribution, fission gas release, creep of fuel and cladding [1, 2]. A thorough description of the time evolution of the isotopic concentration of heavy metals in the fuel is thus crucial. Accordingly, one of the first stages in describing fuel rod behaviour is to compute at each radial position in the fuel: the local burnup, the build-up of heavy metal nuclides and the formation of fission products. The equations used to describe these phenomena constitute the so-called burnup models.

A model called TUBRNP, developed by Lassmann in 1994 [3], is implemented in TU. This model predicts the radial power density distribution based on burnup together with the radial profiles of actinide concentrations.

During the internship, the student will study and verify the implementation of the simplified burnup model. In particular, it will be requested to revise the implementation of the calculation of the radial neutron flux distribution in the fuel slices that is currently preventing the use of user-input distributions.

In addition, to assess the physical consistency of the isotopic composition evolution in case of anomalous behavior, a parallel inventory analysis of LFR fuel rods will be performed using the depletion module of the OpenMC Monte Carlo code.

The student will be introduced to the use of TU for the calculation of fuel rods used in nuclear reactors. He/she will join the team of TU developers at newcleo where he will learn effective collaborative practices and methods for developing code with ensured quality. At the conclusion of the internship, the student is expected to produce a technical report detailing his/her contributions and experiences during the internship period.

Work plan

- a) Literature review and retrieval of the existing work relating to TUBRNP
- b) Code implementation
- c) Analysis of the results
- d) Preparation of a technical report

Applicant profile

Master or PhD student in Nuclear, Industrial or Software Engineering; Applied Physics

Computer skills: FORTRAN, Linux.

Knowledge of $\LaTeX$ editing is appreciated, but not required.

References

- [1] K Lassmann. Transuranus: a fuel rod analysis code ready for use. *Journal of Nuclear Materials*, 188:295–302, 1992.

- [2] K Lassmann and H Blank. Modelling of fuel rod behaviour and recent advances of the transuranus code. *Nuclear Engineering and design*, 106(3):291–313, 1988.
- [3] K. Lassmann, C. O'Carroll, J. van de Laar, and C.T. Walker. The radial distribution of plutonium in high burnup uo2 fuels. *Journal of Nuclear Materials*, 208(3):223–231, 1994.

Refactoring and optimization of DASSH subchannel code

Gabriele Ottino^{*1}, Francesco Pepe^{1,2}, and Daniele Tomatis¹

¹*newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy*

²*Politecnico di Torino, Corso Duca degli Abruzzi 24, 10129 Torino, Italy*

January 14, 2026

– *Internship proposal* –

(ID No. 1009187)

Keywords — LMFR, Subchannel code, Python

Topics — Code development, heat transfer, thermal-hydraulics, numerical simulation

Location newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

Starting date As soon as possible

Duration 6 months

Context

The accurate and fast prediction of thermal-hydraulic behavior of nuclear reactor cores is essential to support and speed up the design activities of Lead-cooled Fast Reactors (LFR). Subchannel codes play a key role in modeling coolant flow, pressure drop, and heat transfer phenomena both in single assemblies and at the core level. Among the others, DASSH ([1], [2], [3]) is a Python-based code originally developed at the Argonne National Laboratory (IL, USA). Currently, newcleo is actively developing it to expand its range of application to include the conditions in which its reactors operate. As the models used in DASSH become more complex, improving its computational performance together with its reliability, maintainability and readability has become a priority. The refactoring of the code is then under evaluation, including the possible use of faster compiled languages like Fortran 2018 or C++ to rewrite some critical sections of the code.

Description

DASSH is a Python-based subchannel code originally used for the steady-state thermal-hydraulic analysis of reactor cores, particularly in liquid metal-cooled fast reactors. The original version of

^{*}Contact person, gabriele.ottino@newcleo.com

the code, developed at the Argonne National Laboratory, is open-source and can be downloaded from <https://github.com/dassh-dev/dassh.git>. *newcleo* has forked the original repository (https://github.com/newcleo-dev-team/dassh_nc) to introduce new advanced features, to improve the code accuracy and to extend its application domain. Although Python ([4], [5]) ensures rapid development and code readability, it could introduce performance limitations when applied to large, computationally demanding problems.

The aim of this internship is to refactor from scratch DASSH code with the aim of improving its readability, maintainability and computational performance. After gathering a deep insight of the current architecture, the candidate will identify the portions of the code that significantly impact the code runtime. Following this diagnostic phase, optimization strategies will be considered and implemented. These may include:

- integration with modules written in C++ or Fortran 2018, that will substitute the computationally most expensive portions of the code;
- adoption of parallelization techniques whenever useful;
- refactoring of some of the already existing Python classes.

During the implementation phase, it will be necessary to ensure that any optimization does not affect the accuracy of the original code, thus updating and/or integrating the test suite accordingly. The refactoring activity will include also the setup of the automatic documentation framework, with the aim of limiting as most as possible any inconsistency between the base code and its manual.

The activity will be carried out under the supervision of the Codes and Methods team at *newcleo*.

Work plan

- a) Analysis of the original code architecture and identification of the most critical sections.
- b) Definition of a more suitable code architecture.
- c) Code implementation.
- d) Update and integration of the test suite, possibly exploiting CI/CD pipelines.
- e) Setup of the automatic documentation framework.
- f) Preparation of a technical report.

Applicant profile

Master student in Industrial Engineering, Computer Engineering or Computer Science.

Skills and background:

- Basic knowledge of object-oriented software programming (Python, Fortran 2018, C/C++);
- Basic knowledge of Linux OS;
- Fundamentals in thermo-fluid dynamics and numerical simulations (appreciated, but not fundamental).

Nice to have: experience using Git and $\text{\LaTeX}$. Basic knowledge of liquid metal-cooled fast reactors.

References

- [1] Milos Atz, Micheal Smith, and Florent Heidet. Dashed software for ducted assembly thermal hydraulics calculations - overview and benchmark. 2020.
- [2] Milos Atz, Micheal A. Smith, and Florent Heidet. *Ducted Assembly Steady State Heat Transfer Software (DASSH) - Theory Manual*. ANL/NSE-21/33, Argonne National Laboratory, 2021.
- [3] Milos Atz, Micheal A. Smith, and Florent Heidet. *Ducted Assembly Steady State Heat Transfer Software (DASSH) - User Guide*. ANL/NSE-21/34, Argonne National Laboratory, 2021.
- [4] Lutz Mark. *Learning Python: Powerful Object-Oriented Programming*. O'Reilly Media, Sebastopol, USA, 5 edition, 2013.
- [5] Giridhar Chetan. *Learning Python Design Patterns*. Packt Publishing, Birmingham - Mumbai, 2 edition, 2016.

Implementation of Helium Contribution to the Fission Gas Swelling and Release Model in the Computer Code TRANSURANUS

Daniele Tomatis^{*1} and Andrea Croci¹

¹newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

January 14, 2026

– *Internship proposal* –
(ID No. 1009581)

Keywords — Helium, Fission Gas Release, Swelling, TRANSURANUS

Topics — nuclear fuel design, fuel performance and thermomechanics, computational modelling

Location newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

Starting date October 2025

Duration 6 months

Context

newcleo is designing an advanced small modular Lead-cooled Fast Reactor (LFR). To perform the fuel performance analysis, newcleo relies on the TRANSURANUS (TU) thermo-mechanical computer code, which is provided by the Joint Research Centre of the European Commission in Karlsruhe, Germany, under a collaboration agreement [1, 2]. newcleo is currently carrying out a verification and validation of the code to check that it can reliably reproduce the physical behaviour of Mixed Oxide (MOX) fuel rods for LFR usage. This ongoing endeavour also includes improving the code to broaden its range of applications, with a specific focus on providing trustworthy predictions for modelling LFR fuel rods. The final goal of this initiative is to develop an improved version of the code.

Description

Among the many phenomena that occur in a fuel pin under irradiation, Fission Gas Release (FGR) and Fission Gas Swelling (FGS) are fundamental to be correctly modelled in a fuel performance code. They are the result of a continuous production of Fission Gas (FG) atoms,

^{*}Contact person, daniele.tomatis@newcleo.com

such as Xenon (Xe) and Krypton (Kr), and other gases like Helium (He), that accumulate in the fuel matrix. This causes volume increase (swelling) and, under certain conditions, such gases can be released into pin's free volume [3] leading to degradation of the thermal conductivity of the fuel rod filling gas (usually He) and increase of the internal fuel rod pressure. These effects reduce the safety margins for the mechanical resistance of the cladding. TU implements a physics-based model to deal with FGR and FGS [4, 5, 6], which, however, does not consider the contribution of He to the gas release and swelling. Commonly employed oxide fuels, especially MOX, produce He through α decay of heavy nuclides and ternary fissions, contributing to the increased internal rod pressure at high burnup levels [7]. In view of its importance for MOX fuels, the inclusion of He in the physics-based model is of great interest.

In this internship, the student will perform a complete literature review of the existing models for FG behaviour, focusing on He and then, he/she will implement in TU all the necessary code changes to include the contribution of He to the swelling and release. Finally, the code will be tested against available experimental data. The student will be introduced to using TU for fuel performance analysis and will become part of the TU development team at newcleo, where they will learn best practices for collaborative development and producing high-quality, reliable code. At the end of the internship, the student is expected to produce a technical report detailing their work and presenting an analysis of the results.

Work plan

- a) Literature review on FG behaviour in oxide and MOX fuels
- b) Code implementation, verification and test
- c) Analysis of the results
- d) Preparation of a technical report

Applicant profile

Master student in Nuclear, Mechanical or Energy Engineering; Applied Physics.

Computer skills: FORTRAN, Linux, Python.

Knowledge of LATEX editing is appreciated, but not required.

References

- [1] K. Lassmann. Transuranus: a fuel rod analysis code ready for use. *Journal of Nuclear Materials*, 188:295–302, 1992.
- [2] K.Lassmann and H. Benk. Modelling of fuel rod behaviour and recent advances of the transurauns code. *Nuclear Engineering and Design*, 106(3):291–313, 1998.
- [3] K. Lassmann and H. Benk. Numerical algorithms for intragranular fission gas release. *Journal of Nuclear Materials*, 280(2):127–135, 2000.
- [4] Giovanni Pastore. *Modelling of Fission Gas Swelling and Release in Oxide Nuclear Fuel and Application to the TRANSURANUS Code*. Doctoral dissertation, Politecnico di Milano, Department of Energy, 2012.

- [5] Giovanni Pastore, Lelio Luzzi, Valentino Di Marcello, and Paul Van Uffelen. Physics-based modelling of fission gas swelling and release in UO_2 applied to integral fuel rod analysis. *Nuclear Engineering and Design*, 256:75–86, 2013.
- [6] Giovanni Pastore, L.P. Swiler, J.D. Hales, S.R. Novascone, D.M. Perez, B.W. Spencer, L. Luzzi, P. Van Uffelen, and R.L. Williamson. Uncertainty and sensitivity analysis of fission gas behavior in engineering-scale fuel modeling. *Journal of Nuclear Materials*, 456:398–408, 2015.
- [7] K. Katsuyama, T. Mitsugi, and T. Asaga. Evaluation of helium release behavior in MOX fuel. In *Proceedings of the 1998 ANS Winter Meeting*, pages 115–116, Washington, D.C., 1998. Materials Science and Technology session.

Numerical study of Fuel Assembly deformations in LFR

Andrea Lario*¹ and Gabriele Ottino¹

¹newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

January 22, 2026

– *Internship proposal* –
(ID No. 1010756)

Keywords — Beam models, FEM, structural deformations

Topics — Structural mechanics, beam theory

Location newcleo S.p.A., Corso Stati Uniti 38, 10128, Torino, Italy

Starting date Second quarter of 2026

Duration 5–6 months

Context

To design an inherently safe Lead-cooled Fast Reactor (LFR), it is crucial to predict mechanical deformations accurately since changes in the spacing between neighbouring fuel assemblies (FAs) can result in altered neutron production and absorption, thus affecting the overall reactivity. Accurate prediction of FA deformations can be achieved by using structural computational tools based on Finite Element Methods (FEMs). However, their intrinsically high computational cost and the need to perform many calculations to solve the coupling between neutronics and thermal-hydraulics limit their application in ordinary design work. An alternative approach is based on simplified 1D methods, which approximate each FA as a beam element. While this is a considerable simplification, when applied to nuclear core reactors it retains most of the physical phenomena that are of interest for investigation and ensures fast predictions, requiring a limited number of input parameters. There are only a few scientific tools available for analysing FA deformation in core reactors that are based on 1D tools [1, 2]. The goal is to develop a scientific computational tool based on 1D beam elements for design and safety calculations, accepting some degree of approximation in the solution.

*Contact person: andrea.lario@newcleo.com

Description

Both elastic and inelastic deformations affect the shape of fuel assemblies in a reactor core. These deformations are driven by various factors; in particular, high thermal gradients induce bowing, while irradiation and stress induce the inelastic effects of creep and swelling. Additionally, local forces acting on an assembly can arise when FAs come into contact with each other.

The aim of the internship is to investigate the deformation of fuel assemblies when subjected to elastic and inelastic deformations. The investigation will be carried out by using models of different accuracy and detail. After gaining an in-depth understanding of the existing literature, the candidate will identify the most suitable models and range of parameters for describing the configurations of interest at *newcleo* in terms of, for example, geometry, temperature gradients, irradiation and materials.

Beam models based on Bernoulli or Timoshenko theory will be implemented, and their predictions will be compared with analytical solutions (whenever available) or results produced with FEM solvers for structural mechanics.

Expected results include: a comparison between the considered models; the identification of the parameters that most affect the deformation of fuel assemblies; the evaluation of modelling the single phenomena involved; the study of the mutual interaction among these phenomena.

This work will be carried out under the supervision of the Codes and Methods team at *newcleo*.

Work plan

- (a) Provide a comprehensive review of the literature;
- (b) Identify the relevant physical phenomena and aspects of *newcleo*'s LFR core and fuel elements;
- (c) Select test configurations and compare the results provided by models of different accuracy;
- (d) Perform parametric studies to identify the factors that have the greatest influence on the results and to assess the effects introduced by simplifications;
- (e) Prepare a technical report.

Applicant profile

Master student in Nuclear Engineering, Civil Engineering, Aerospace Engineering or Mechanical Engineering. Skills and background:

- Basic knowledge of software programming (Python/FORTRAN 2018 are appreciated);
- Basic knowledge of Linux OS;
- Fundamentals in numerical simulations;
- Knowledge of Finite Element Method software for structural mechanics, either commercial or open-source.

Nice to have: experience using $\text{\LaTeX}$. Basic knowledge of liquid metal-cooled fast reactors.

References

- [1] Nicholas Wozniak, Emily R. Shemon, and James J. Grudzinski. Review of tools for modeling core radial expansion in liquid metal-cooled fast reactors. Technical report, Argonne National Laboratory (ANL), Argonne, IL (United States), Nov. 2020.
- [2] G. A. McLennan NUBOW: a FORTRAN-IV program for the static elastic structural analysis of bowed reactor cores Technical report, Argonne National Laboratory (ANL), Argonne, IL (United States), Apr. 1974.

References

Beam-to-beam contacts for modelling Fuel Assembly interactions in LFR

Andrea Lario^{*1} and Gabriele Ottino¹

¹*newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy*

January 26, 2026

– *Internship proposal* –
(ID No. 1010757)

Keywords — Contact model, structural deformations, numerical modelling

Topics — Structural mechanics, beam theory

Location Corso Stati Uniti 38, 10128, Torino, Italy

Starting date Second quarter of 2026

Duration 5–6 months

Context

Mechanical deformation of the core structures causes relevant reactivity feedback in fast reactors. Accurate prediction of the geometry modifications and of their positioning into the core can be obtained by using structural computational tools based on Finite Element Methods (FEMs). However, the high computational cost of these tools and the need to perform many calculations to solve the coupling between neutronics and thermal-hydraulics limit their application in ordinary design work. To overcome these limitations, simplified mechanical models with lower computational effort should be used. These approaches, based on 1D beam theory, trade computational speeds for accuracy, but they still describe the phenomena (thermal bowing, creep, swelling, and contacts) which contribute the most to the mechanical deformation of the FAs. There are only a few scientific tools available for analysing FA deformation in core reactors that are based on 1D tools [1, 2].

The goal of the research is to develop a scientific computational tool that can predict fast the deformation of fuel assemblies by considering the reciprocal forces acting on the contacting beams.

^{*}Contact person andrea.lario@newcleo.com

Description

Significant deformations due to both elastic and inelastic effects can result in bowed FAs coming into contact with each other during the operative life of the fuel assembly.

The fact that fuel assemblies can either be in contact or have clearances introduces non-linearities in the system of equations which describe the mechanical behaviour of the core, making ordinary linear mathematical techniques inapplicable. On the other hand, iterative solvers could be computationally expensive or lack stability.

The internship aims at studying an efficient and robust numerical techniques for solving contacts among FAs approximated as 1D beams.

After gathering a deep insight of the works available in the literature identifying appropriate test cases, the candidate will select suitable methods to solve the contacts efficiently and robustly. Two distinct models shall be possibly developed:

- simplified 2D contact model, which will be used to predict the deformation of FAs when they mutually come into contact;
- possible extension of the model to 3D geometry.

In both cases results will be compared against data available in the literature or obtained using FEM-based codes for structural analysis. Codes will be written in both Fortran 2018 and Python.

The activity will be carried out under the supervision of the *Codes and Methods* team at newcleo.

Work plan

- (a) Provide a comprehensive literature review;
- (b) Select the best methods for solving contacts occurring when beam elements come in contacts;
- (c) Identify proper test cases;
- (d) Implement and verify the selected methods, focusing on numerical robustness and efficiency;
- (e) Prepare a technical report.

Applicant profile

Master student in Nuclear Engineering, Civil Engineering, Aerospace Engineering or Mechanical Engineering. Skills and background:

- Basic knowledge of software programming (Python/FORTRAN 2018 are appreciated);
- Basic knowledge of Linux OS;
- Fundamentals in numerical simulations and numerical analysis.

Nice to have: experience using Git and $\text{\LaTeX}$. Basic knowledge of liquid metal-cooled fast reactors.

References

- [1] Nicholas Wozniak, Emily R. Shemon, and James J. Grudzinski. Review of tools for modeling core radial expansion in liquid metal-cooled fast reactors. Technical report, Argonne National Laboratory (ANL), Argonne, IL (United States), Nov. 2020.
- [2] G. A. McLennan NUBOW: a FORTRAN-IV program for the static elastic structural analysis of bowed reactor cores Technical report, Argonne National Laboratory (ANL), Argonne, IL (United States), Apr. 1974.

References

Full-core calculation of LFR for maritime applications

Gianluca Cirillo¹ and Daniele Tomatis ^{*1}

¹newcleo, Corso Stati Uniti 38, 10128 Torino, Italy

December 12, 2025

– *Internship proposal* –
(ID No. 1010758)

Keywords — LFR, DRAGON-DONJON, deterministic neutron transport, core cycle.

Topics — Reactor physics, core analysis and design, fuel management.

Location Turin office, Italy or Saclay office, France

Starting date Beginning of 2026

Duration 6 months

Context

newcleo is developing innovative small modular reactors based on lead-cooled fast reactor (LFR) technology. A sealed unit intended for long-term extended operation is currently under study for ship propulsion and remote/off-shore power generation. The internship focuses on the setup of an advanced calculation scheme for this LFR unit by the deterministic computer codes DRAGON5/DONJON [1], with comparison against the Monte Carlo computer code OpenMC [2] for performance assessment.

Description

The reactor considered in this work is of the sealed type seeking inherent safety by means of passive cooling systems and employing robust materials to ensure long-term reliability, as in [3, 4]. Continuous reactor operation for at least 20 years is targeted. Excess reactivity control at beginning of cycle is achieved by means of fertile elements and burnable absorbers.

The internship focuses on full-core neutron calculations of liquid lead fast reactors (LFRs) using the deterministic codes DRAGON/DONJON. Calculations are in eigenvalue mode to reproduce operation in critical state by control elements progressively inserted over time. The feasibility of using different types of control elements will be investigated. The calculations will

*Contact person, daniele.tomatis@newcleo.com

be performed with assumed distributions of temperature and thermal expansion for geometry deformation.

Several calculations are necessary to find optimal configurations for reactor design, which limits the applicability of the Monte Carlo method indeed. The use of fast deterministic methods is then essential in the engineering framework.

During the internship, the student will be introduced to the physics and engineering of the LFR reactors, acquiring notions and developing skills in deterministic transport methods and neutral particle transport [5]. The student will become a proficient user of DRAGON/DONJON and will improve his/her Python programming to perform post-processing and analysis of simulation results. At the end of the internship, the student will write a technical report on the analysis of the results from the numerical simulations. The student will be followed by members of the Codes and Methods group of *newcleo*.

Work plan

During the internship, the student will be introduced to open-source transport computer codes for modeling LFR reactors and calculating maps of the released thermal power.

The student will contribute to the setup of an industrial calculation scheme suitable for fast reactor physics. These include the theoretical aspects underlying particle transport simulation and adequate modeling of the fuel elements.

The expected working plan is detailed in the following:

- a) Literature review on off-grid and long-term reactors
- b) Discussion of target key performance parameters
- c) DRAGON/DONJON introduction and training
- d) Setup of the LFR full reactor model
- e) Analysis of the results
- f) Preparation of a technical report

Applicant profile

Master or PhD student in Nuclear Engineering, Applied Mathematics and Physics.

Background: fundamentals in reactor physics and Monte Carlo methods.

Required computer skills: A good knowledge of Python and Linux operating systems (or willingness to learn). Knowledge of $\text{\LaTeX}$ editing is appreciated.

References

- [1] Guy Marleau, Alain Hébert, and Robert Roy. A user guide for dragon version 5, 2025.
- [2] Paul K. Romano, Nicholas E. Horelik, Bryan R. Herman, Adam G. Nelson, Benoit Forget, and Kord Smith. Openmc: A state-of-the-art monte carlo code for research and development. *Annals of Nuclear Energy*, 82:90–97, 2015. Joint International Conference on Supercomputing in Nuclear Applications and Monte Carlo 2013, SNA + MC 2013. Pluri- and Trans-disciplinarity, Towards New Modeling and Numerical Simulation Paradigms.

- [3] J. Wallenius, S. Qvist, I. Mickus, S. Bortot, P. Szakalos, and J. Ejenstam. Design of sealer, a very small lead-cooled reactor for commercial power production in off-grid applications. *Nuclear Engineering and Design*, 338:23–33, 2018.
- [4] Keith E Holbert. A review of maritime nuclear reactor systems. *Journal of Nuclear Engineering*, 6(1):5, 2025.
- [5] George I Bell and Samuel Glasstone. *Nuclear reactor theory*. US Atomic Energy Commission, Washington, DC (United States), 1970.
- [6] Alain Hébert. *Applied reactor physics*. Presses inter Polytechnique, 2009.

Development and Validation of an Inherent Neutron Source Code

Matteo Zammataro ^{*1}, Ilaria Calsalbore², and Sigtryggur Hauksson³

^{1,2}newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

³newcleo SA, EDF Lab Paris-Saclay, Bd Gaspard Monge, 91120 Palaiseau, France

January 14, 2026

– *Internship proposal* –
(ID No. 1011110)

Keywords — Inherent neutron sources, (α, n) reactions, spontaneous fissions

Topics — Particle physics, code development, verification & validation

Location newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

Starting date First/second quarter 2026

Duration 5–6 months

Context

newcleo is developing a new generation of small modular lead-cooled fast reactors (LFRs), with the goal of improving fuel utilization, reducing high-level waste, and enabling closed fuel cycles. In this context, it is critical to have reliable, quantitative data about all inherent neutron sources in fuels under design or use, since these sources have an impact on safety and radiation protection, especially under accident or handling scenarios.

Inherent neutron sources typically arise from spontaneous fission and (α, n) reactions, with a smaller contribution from delayed neutron emission. Existing computational tools for estimating these sources, such as [2], exhibit limitations in flexibility, data-library compatibility, and they are not validated for fast-spectrum reactor applications.

This internship focuses on the development of an alternative computational tool for inherent neutron sources calculation and on the identification and analysis of experimental data suitable for its validation.

*Contact person, matteo.zammataro@newcleo.com

Description

The design of a new reactor unit requires a clear and reliable assessment of the inherent neutron sources generated within the nuclear fuel under different irradiation conditions. These sources influence every phase of reactor life: during start-up their characterization is needed to size and justify the external start-up neutron source; during shutdown, spontaneous emissions meaningfully contribute to the residual neutron flux; and in post-irradiation fuel handling and disposal, accurate neutron source assessments are crucial for radiation protection evaluations.

Inherent neutron sources arise mainly from three phenomena: spontaneous fission, (α, n) reactions and, marginally, delayed neutron emission [1]. While contributions related to delayed neutron emission are typically negligible, spontaneous fission and (α, n) reactions provide source terms that cannot be ignored.

Although scientific codes exist to estimate these neutron sources, most show limitations in structure, robustness, and compatibility with modern nuclear data libraries. Furthermore, no dedicated validation effort has yet been carried out for fast-reactor applications.

In this internship, the student will contribute to the development of a scientific computer tool for the calculation of inherent neutron sources. The work will also involve identifying and analyzing experimental data to support the validation of the developed code. As part of the activity, the student will review the state-of-the-art tools that already exist [2], identify their limitations, and implement strategic improvements. This will provide hands-on experience with programming languages like Fortran and Python, exposure to numerical methods, and allow the student to become familiar with modern software-development good practices. The results of this work will improve the robustness of inherent neutron source calculations, improving confidence in neutronics, safety, and radiation-protection analyses. The internship will be carried out under the supervision of the Codes & Methods group at newcleo.

Work plan

The internship will involve both theoretical and practical work, combining the study of the underlying physics of α particles transport, spontaneous fissions and (α, n) reactions with applied numerical methods and scientific programming. The student will learn how to use modern DevOps platforms, such as Git, and improve his knowledge of Fortran and Python.

The expected working plan is as follows:

- a) Literature review and study of the main physical phenomena.
- b) Training with SOURCES4C.
- c) Identification of the limitations and discussion of possible improvements.
- d) Contribution to the development of a new tool for inherent neutron sources calculation.
- e) Identification of experimental data for code validation.
- f) Preparation of the final report.

Applicant profile

Master student in Nuclear Engineering or student in Applied Mathematics and Physics.

Background: fundamentals in nuclear physics and applied mathematics.

Required computer skills: Linux, Python (intermediate), Fortran (nice to have) $\text{\LaTeX}$ scientific editing (optional).

References

- [1] Rui-Min Ji, Cheng-Gang Yu, Ming-Hai Li, Rui Yan, Yang Zou, and Gui-Min Liu. Study on inherent neutron sources in MSR. *Nuclear Science and Techniques*, 29:47, 2018.
- [2] W.B. Wilson, R.T. Perry, W.S. Charlton, and T.A. Parish. Sources: A code for calculating (α , n), spontaneous fission, and delayed neutron sources and spectra. *Progress in Nuclear Energy*, 51(4):608–613, 2009.

Chemistry on molten lead impurities behaviour to support LFR design and licensing

– *Internship Proposal* –

(ID No. 1011150)

Luca Silvioli^{*1} and Francisco Garcia Ferre²

¹ newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

²newcleo S.p.A., Via Galliano 27, 10129 Torino (TO), Italia

March 19th, 2025

Keywords — Chemistry control, oxygen sensors, molten metals, EOP, LFR, nuclear, electrochemistry, elemental analysis

Topics — electrochemistry, chemical thermodynamics, corrosion, analytical chemistry, inorganic chemistry, corrosion science

Location newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

Starting date To be defined

Duration 6 months

Context

Lead-cooled fast reactors (LFRs) are considered one of the most promising reactors among several advanced nuclear systems. The operation of small modular reactors with a fast neutron spectrum poses several material challenges to scientists and engineers due to the extreme environmental conditions inside the reactor[1]. Therefore, the understanding of the reactor chemical systems is key to successfully deploy new power reactors [2]. *newcleo* is committed to a large R&D programme to mature chemistry knowledge and support the design of chemical unit processes and operations for nuclear applications. In this framework, the aim of the internship is to support *newcleo* chemists in understanding the chemical behavior of various chemical species in molten lead and providing support during and after experimental campaigns, either by performing the experiments, or carrying on the post-test analysis.

* Contact person: luca.silvioli@newcleo.com

Description

Liquid lead is the chosen coolant for the LFR. The understanding and monitoring of impurities dissolved in the coolant is a crucial requirement, influencing the oxidation of the containment materials, the mobility of chemical species across the reactor and ultimately impact the design and the licensing process. At *newcleo Ltd.*, to support our LFR-AS-30 programme, we have established a new chemistry laboratory where to study impurities behaviour (activity, solubility, diffusion) and their speciation in molten lead. Combining electrochemical methods with elemental analysis and microscopy, and using custom made test rigs, we seek to understand how the coolant chemistry is influenced by the reactor process parameters, and how such chemistry impact the functioning of the reactor control and safety systems.

The successful candidate will review the scientific literature and internal documentation to create a suitable test matrix. They will learn how to use custom, state of the art sensing equipment for molten lead, and work in Brasimone laboratories where they can contribute to the commission and operation experimental equipment. The candidate will further carry out testing and data processing and interpretation, learning how to present clearly their findings in a scientific report of outstanding quality to meet regulatory standards.

The intern may also work in collaboration with the *newcleo's* branches in Turin (where other equipment is available) and Lyon (where engineering studies are carried out).

Work plan

- a) Literature review & project scoping
- b) Test matrix & experimental specifications creation
- c) Experimental testing with custom equipment
- d) Data analysis and interpretation
- e) Preparation of reports and final presentation

Applicant profile

Master student in Chemistry, Industrial Chemistry, Applied Physics, Chemical Engineering, or equivalent.

Skills:

- Chemical thermodynamics
- Chemical unit operations and processes
- Analytical chemistry techniques, with focus on electrochemical methods
- Hands-on expertise in general chemistry lab.
- Experience with programming languages (Matlab, Python) is a plus.
- Languages: English (at least B2 level). French is a plus.

References

- [1]. Malerba, L. et al., “Materials for Sustainable Nuclear Energy: A European Strategic Research and Innovation Agenda for All Reactor Generations”, *Energies*, 15, 1845 (2022)
- [2]. Aerts, K. Gladinez, B. G. Prieto, J. Lim, A. Marino, K. Rosseel, “The LBE Coolant Chemistry R&D Programme for the MYRRHA ADS: Chemistry and Control of Oxygen, Corrosion and Spallation Products”, *Proceedings of the 14th International Workshop on Spallation Materials Technology, JPS Conference Proceedings*, 28 (2020)

Material characterization after corrosion tests in static and flowing lead

– *Internship Proposal* –

(ID No. 1011151)

Luca Silvioli^{*1} and Francisco Garcia Ferre²

¹ newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

² newcleo S.p.A., Via Galliano 27, 10129 Torino (TO), Italia

August 1st, 2025

Keywords — LFR materials, corrosion, metals, coatings

Topics — stainless steel, optical microscopy, mechanical tests, secondary electron microscopy, x-ray diffraction, surface treatments

Location newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

Starting date To be defined

Duration 6 months

Context

Lead-cooled fast reactors (LFRs) are considered one of the most promising reactors among several advanced nuclear systems. Qualification and standardization of structural and core materials are fundamental aspects of the newcleo project. Given the highly aggressive environment of Lead-cooled Fast Reactors (LFRs), material selection must meet multiple criteria - most notably, resistance to heavy metal corrosion and liquid metal embrittlement. Demonstrating these properties is a prerequisite for submitting material data to regulatory authorities as part of the reactor prototype licensing process. In this framework, the aim of the internship is to study the current state of the art of nuclear structural materials, to support commissioning and upgrades of experimental equipment and to carry out experimental tests, producing a report that analyses the generated experimental data.

* Contact person: luca.silvioli@newcleo.com

Description

Corrosion protection of structural materials in Lead-cooled Fast Reactors (LFRs) remains one of the primary challenges limiting the development of this advanced nuclear technology. To address this, **newcleo** has established dedicated experimental facilities at its Brasimone site to investigate material performance in liquid lead environments.

Two key installations - **CAPSULE** (for static liquid Pb) and **CORE** (for circulating Pb) - are specifically designed to study the corrosion and erosion behavior of candidate materials. Understanding the mechanisms leading to material degradation in LFR conditions is essential for improving reactor safety and advancing the development of next-generation nuclear materials.

The selected intern will begin by reviewing relevant scientific literature and internal documentation to gain a comprehensive understanding of the current state of the art. They will contribute to the enhancement of experimental setups at the Brasimone laboratories and assist in the preparation and characterization of material samples using one or more of the following techniques:

- Optical Microscopy
- Scanning Electron Microscopy (SEM) with EDX and EBSD
- X-ray Diffraction (XRD)

In addition, the intern will collaborate with **newcleo** teams in **Turin**—where additional instruments such as high-temperature nanoindentation are available—and **Lyon**, where related engineering analyses are conducted.

Work plan

- a) Literature review & materials selection
- b) Experimental campaign planning
- c) Materials testing with custom equipment
- d) Data analysis
- e) Preparation of reports and final presentation

Applicant profile

Master student in Nuclear Engineering, Mechanical Engineering, Materials Science, Chemistry, Physics, Chemical Engineering, or equivalent.

Skills:

- Knowledge of material manufacturing processes, testing and characterization techniques.

- Required computer skills: knowledge of Excel and data analysis software. Experience with programming languages (Matlab, Python) is a plus.
- Languages: English (at least B2 level). French is a plus.

References

- [1] Malerba, L. et al., Materials for Sustainable Nuclear Energy: A European Strategic Research and Innovation Agenda for All Reactor Generations. *Energies*, **15**, 1845 (2022)
- [2] Fulger, M., Khumsa-Ang, K., Šípová, M., Ducu, C. M. & Sáez-Maderuelo, A. Special Issue: Behavior of Materials (Alloys, Coatings) in Conditions Specific to Gen IV Nuclear Reactors., *Coatings*. **13**, 58 (2022)

Oxygen control studies on lead coolant for fast reactor nuclear systems

– *Internship Proposal* –

(ID No. 1011152)

Luca Silvioli¹ and Francisco Garcia Ferre²

¹ newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

² newcleo S.p.A., Via Galliano 27, 10129 Torino (TO), Italia

March 19th, 2025

Keywords — Chemistry control, oxygen sensors, molten metals, EOP, LFR, nuclear, electrochemistry, elemental analysis

Topics — electrochemistry, chemical thermodynamics, corrosion, analytical chemistry, inorganic chemistry

Location newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

Starting date To be defined

Duration 6 months

Context

Lead-cooled Fast Reactor (LFR) are considered one of the most promising reactors among several advanced nuclear systems. The operation of small modular reactors with a fast neutron spectrum poses several chemical and material challenges to scientists and engineers due to the exotic environmental conditions inside the reactor [1]. The understanding of the reactor coolant chemistry behavior is particularly key to successfully deploy these new reactor technology [2]. newcleo is committed to a large R&D programme to mature chemistry knowledge and support the design of chemical unit processes and operations for nuclear applications. In this framework, the aim of the internship is to support newcleo chemists in understanding the chemical reactivity and mobility of oxygen in the coolant and providing support during and after experimental campaigns, either by performing the experiments, or carrying on the post-test analysis

* Contact person: luca.silvioli@newcleo.com

Description

Liquid lead is the chosen coolant for the LFR. The control of oxygen dissolved in the coolant is a crucial requirement, influencing the oxidation of the steels, dissolved corrosion products and lead itself. At *newcleo ltd.*, to support our LFR-AS-30 programme, in collaboration with ENEA we routinely use our technology for the Electrochemical Oxygen Sensors (EOSs) and Pumps (EOPs) to shed light on the O-Pb-PbO chemical system in molten lead. Specifically, combining potentiometry by the EOSs with the coulometry by the EOPs along with a customized setup, we aim to fully describe how process parameter variables affect the oxygen activity in lead, to yield a robust chemistry control strategy suitable to scale-up.

The successful candidate will review the scientific literature and internal documentation to create a suitable test matrix. They will learn how to use custom, state of the art sensing equipment for molten lead, specifically the EOS and EOP technologies, and work in Brasimone laboratories where they can contribute to the commission and operation experimental equipment. The candidate will further carry out testing and data processing and interpretation, learning how to present clearly their findings in a scientific report of outstanding quality to meet regulatory standards.

The intern may also work in collaboration with the *newcleo's* branches in Turin (where other equipment is available) and Lyon (where engineering studies are carried out).

Work plan

- a) Literature review & project scoping
- b) Test matrix & experimental specifications creation
- c) Experimental testing with custom equipment
- d) Data analysis and interpretation
- e) Preparation of reports and final presentation

Applicant profile

Master student in Chemistry, Industrial Chemistry, Applied Physics, Chemical Engineering, or equivalent.

Skills:

- Chemical thermodynamics
- Chemical unit operations and processes
- Analytical chemistry techniques, with focus on electrochemical methods
- Hands-on expertise in general chemistry lab.
- Experience with programming languages (Matlab, Python) is a plus.
- Languages: English (at least B2 level). French is a plus.

References

- [1]. Malerba, L. et al., “Materials for Sustainable Nuclear Energy: A European Strategic Research and Innovation Agenda for All Reactor Generations”, *Energies*, 15, 1845 (2022)
- [2]. Aerts, K. Gladinez, B. G. Prieto, J. Lim, A. Marino, K. Rosseel, “The LBE Coolant Chemistry R&D Programme for the MYRRHA ADS: Chemistry and Control of Oxygen, Corrosion and Spallation Products”, *Proceedings of the 14th International Workshop on Spallation Materials Technology, JPS Conference Proceedings*, 28 (2020)

Testing and characterization of Pb retained on structural materials for advanced nuclear systems

– *Internship Proposal* –

(ID No. 1011153)

Luca Silvioli^{*1} and Francisco Garcia Ferre²

¹ newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

² newcleo S.p.A., Via Galliano 27, 10129 Torino (TO), Italia

March 19th, 2025

Keywords — LFR materials, corrosion, metals, coatings, Pb retention

Topics — stainless steel, optical microscopy, corrosion tests, secondary electron microscopy, x-ray diffraction, surface treatments

Location newcleo S.p.A. C.R. ENEA Brasimone 40032 Camugnano (BO), Italia

Starting date To be defined

Duration 6 months

Context

Lead-cooled fast reactors (LFRs) are considered one of the most promising reactors among several advanced nuclear systems. The operation of small modular reactors with a fast neutron spectrum poses several material challenges to scientists and engineers due to the extreme environmental conditions inside the reactor [1]. Therefore, the development of advanced materials is fundamental to successfully deploy new power reactors [2]. newcleo is committed to a large RD programme to assess the feasibility of metallic materials and coatings for nuclear applications. In this framework, the aim of the internship is to study the current state of the art of nuclear structural materials, investigate Pb retention on different materials, and providing support during and after experimental campaigns, either by performing the experiments, or carrying on the post test analysis.

* Contact person: luca.silvioli@newcleo.com

Description

Pb retention on materials used in LFRs is one of the issues that hinders the development of this class of nuclear reactors. Pb retention represents an issue when components inside the LFR need to be moved (i.e. when fuel assemblies are lifted during refuelling, inspection equipment and monitoring devices being removed from the reactor vessel), as it might cause spread of activated elements (i.e. Po formed from Pb activation), contamination of the CIS (Component Isolation System) with Pb residuals which needs to be accounted during CIS's design phase.

Understanding which parameters affect Pb retention on reactor components plays a vital role in improving the safety criteria of nuclear reactors and promoting research on advanced materials.

The successful candidate will review the scientific literature and internal documentation to select suitable materials for testing. The intern will work in Brasimone laboratories to commission experimental equipment and prepare materials specimens to characterize through one or more of the following techniques:

- Optical microscopy
- Scanning Electron Microscopy equipped with EDX and EBSD
- X-ray diffraction
- Corrosion tests
- Other tests in collaboration with *newcleo*'s branch in Turin

The intern will also work in collaboration with the *newcleo*'s branches in Turin (where other equipment will be available, such as (high temperature) nanoindentation) and Lyon (where engineering studies are carried out).

Work plan

- a) Literature review and materials selection
- b) Experimental campaign planning
- c) Materials testing with custom equipment
- d) Data analysis
- e) Preparation of reports and final presentation

Applicant profile

Master student in Nuclear Engineering, Mechanical Engineering, Materials Science, Chemistry, Physics, Chemical Engineering, or equivalent.

Skills:

- Knowledge of material manufacturing processes, testing and characterization techniques.
- Required computer skills: knowledge of Excel and data analysis software. Experience with programming languages (Matlab, Python) is a plus.

- Languages: English (at least B2 level). French is a plus.

References

- [1]. Fulger, M., Khumsa-Ang, K., Šípová, M., Ducu, C. M. and Sáez-Maderuelo. (2022). A. Special Issue: Behavior of Materials (Alloys, Coatings) in Conditions Specific to Gen IV Nuclear Reactors. *Coatings.*, 13(58).
- [2]. Malerba, L. et al. (2022). Materials for Sustainable Nuclear Energy: A European Strategic Research and Innovation Agenda for All Reactor Generations. *Energies*, 15(1845).

Parallelization of neutron transport calculations in Dragon5.1

Alain Hébert¹, Sigtryggur Hauksson * ¹, and Daniele Tomatis²

¹*newcleo, EDF Lab Paris-Saclay, Bd Gaspard Monge, 91120 Palaiseau, France*

²*newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy*

January 26, 2026

– *Internship proposal* –

(ID No. 1011283)

Keywords — Lattice calculation, DRAGON, GPUs, OpenMP

Topics — Neutron transport, parallel computing, computer science

Location EDF Lab Paris-Saclay, Bd Gaspard Monge, 91120 Palaiseau, or *newcleo* S.p.A., Via Galliano 27, 10129 Torino, Italy

Starting date Spring 2026

Duration 6 months

Context

newcleo is designing new Lead-cooled Fast Reactor (LFR) units for small modular reactors. These reactors sustain a fast neutron spectrum allowing for fission of heavier actinides, such as plutonium that is a by-product of light water reactors. Cooling is done by molten lead which shields photon radiation and has low neutron absorption.

The design and operation of reactors requires fast transport calculations that solve the Boltzmann equation for neutrons in the core of the reactor. Deterministic transport computer codes of choice are DRAGON5, which performs lattice calculations, and the full-core code DONJON [2, 1]. Both of these codes have been developed at Polytechnique Montréal, Québec, Canada, and are used at *newcleo* for LFR calculations. They are written in Fortran-2003. The goal of this internship is to speed up calculations with DRAGON5, focusing on calculations with the discrete ordinates (S_N method) or the method of characteristics (MOC) which both serve to solve the transport equation in multigroup approximation. This student will analyze the code, and suggest and implement improvements in parallelization implemented on multi-core processors, multiple processing units or graphical processing units.

*Contact person, sigtryggur.hauksson@newcleo.com

Description

In the first part of the internship, the student will analyze in detail the performance and the structure of the code to be sped up. This will either be the code implementing the S_N or the MOC method, both of which already have a parallelization in OpenMP. The analysis of the code will be helped by presence of developers of Dragon in newcleo's team. The student will then, in consultation with the supervisors, propose which tools will be best adapted to improve performance. A non-exhaustive list of options are:

1. Improved use of OpenMP pragmas.
2. Distributed parallelism with MPI for calculation across multiple processing units.
3. Implementation on graphical processing units (GPUs) with CUDA or OpenACC.
4. Coarray processing following the Fortran2018 specification.

This choice will take into account calculational facilities at newcleo. In the second part of the internship, the student will develop the details of the calculation scheme proposed, testing different options to maximize speedup. Results of the internship will be written up in a final report.

Work plan

- a) Analysis of the structure of the relevant modules in Dragon.
- b) Proposal for architecture to perform parallelization on.
- c) Design of parallelized algorithm.
- d) Implementation, testing of different options, measures of speedup.
- e) Preparation of the final report.

Applicant profile

Master student in Computer Science.

Background: Experience with parallelization in scientific computing is preferable.

Required computer skills: Fortran, OpenMP, CUDA or OpenACC, $\text{\LaTeX}$ scientific editing (optional).

References

- [1] Alain Hébert. DRAGON5 and DONJON5, the contribution of École Polytechnique de Montréal to the SALOME platform. *Annals of Nuclear Energy*, 87:12–20, 2016.
- [2] G. Marleau, A. Hébert, and R. Roy. A user guide for DRAGON Version 5, 2025.

Innovative materials for reactor control and shielding in small modular LFR design

Alex Aimetta* ¹ and Daniele Tomatis¹

¹*newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy*

February 3, 2026

– *Internship proposal* –
(ID No. 1011320)

Keywords — LFR, Monte Carlo, core cycle, neutron shielding, neutron activation

Topics — Reactor physics and analysis, radiation transport, core design.

Location newcleo S.p.A., Via Galliano 27, 10129 Torino, Italy

Starting date To be defined

Duration 6 months

Context

newcleo is developing innovative small modular reactors based on lead-cooled fast reactor (LFR) technology. The use of alternative materials is considered to satisfy the main aim of these reactors: the achievement of target performance for long-term operation without refueling. Among the different alternative materials that can be used in small modular LFR, beryllium could be a promising candidate, thanks to its expected capability of reducing the critical size of nuclear systems. Moreover, in this type of compact fast reactors, shielding components from high neutron fluxes coming from the core is of primary importance in order to avoid phenomena related to neutron activation and material degradation. This internship focuses on performing a complete literature review about the utilization of innovative materials (e.g., hafnium, erbium and beryllium) in nuclear reactors, providing the engineering teams with essential data to improve the design of future reactor units. Then, computations with Monte Carlo (MC) codes will be performed by the intern to assess how the use of these materials in newcleo's designs can be beneficial and to determine which core configuration maximizes the benefits of employing them in the reactor.

*Contact person, alex.aimetta@newcleo.com

Description

The results provided by this internship will support the design of a small modular LFR, where the employment of beryllium, thanks to $(n,2n)$ reactions could be particularly beneficial in reducing the size of the systems, when combined with innovative shielding materials.

The internship will start with a bibliographic review about the possibility of employing materials like erbium and hafnium for the neutron shielding and beryllium as a reflector in small modular LFR. In the literature, there are studies which show the impact in terms of reactivity of the $(n,2n)$ reactions of beryllium, leading to systems with smaller critical dimensions [1], which is a particularly important characteristic for small modular LFR [2]. However, other studies show that beryllium is affected by severe issues in terms of neutron activation and gas production, making the disposal of systems employing beryllium an even more delicate task [3]. Additionally, beryllium is characterized by quite high values of (γ,n) cross sections at energies characteristic of fission product's γ radiation: it has been shown that the impact of photonuclear neutrons can be neglected for stationary states, but it has non-negligible effects on the kinetics of the reactor [4, 5]. The presence of lead, which has a strong shielding effect against photons, could possibly alleviate the magnitude of these phenomena.

The expected increase of the neutron flux due to the presence of beryllium requires a detailed study of shielding configurations. In particular, the importance of analysing alternative shielding materials lies in the fact that boron carbide, the best one in terms of neutron absorption, has many drawbacks like swelling and loss of performance under irradiation. In this sense, the employment of more stable materials like hafnium [6] and erbium [7] can be advantageous, in particular when the aim is to achieve long-term operation. On the other hand, the main drawbacks of Hf and Er are their lower absorption efficiency and their high photon production after radiative capture interactions, which, however, could be compensated by the presence of lead.

The student will have to find several references in the literature in order to understand if the advantages of using innovative materials in fast fission reactors are supported by physical and engineering considerations, taking into account all the pros and the cons of employing such nuclides and analysing their behaviour in the presence of lead and of a fast spectrum. This phase will be followed by computational evaluations with Monte Carlo codes like OpenMC [8], to quantitatively determine the impact of adding erbium, hafnium and beryllium to *newcleo*'s reactors using different geometry configurations and finding the optimal one in terms of nuclear performance. Integral quantities (e.g., multiplication factor and kinetic parameters), neutron fluxes and spectra and burnup will be computed, investigating the main variations induced by the presence of innovative materials on these responses.

During the internship, the student will have to analyse the information and the results retrieved during the bibliographic review and thanks to the neutronic computations, with the aim of providing valuable information about the possible use of innovative materials for the core control and neutron shielding in the design of a small modular LFR, supporting the design of this nuclear system. At the end of the internship, the student will write a technical report summarizing all the main findings of the activity. The student will be followed by the Codes and Methods group members of *newcleo*.

Work plan

During the internship, the student will need to perform a detailed bibliographic review, accompanied by high-fidelity MC calculations, in order to conclude how the design of small modular LFR

can benefit from the use of innovative materials like Er, Hf and Be.

The expected working plan is detailed in the following:

- a) Literature review on erbium, hafnium and beryllium behaviour in fast systems
- b) Literature review on fission systems employing erbium, hafnium and beryllium
- c) Discussion of current core configurations
- d) Analysis of bibliographic references
- e) Neutronic calculations with MC codes
- f) Preparation of a technical report

Applicant profile

Bachelor/Master student in Nuclear Engineering, Applied Mathematics and Physics.

Background: fundamentals in reactor physics.

Required computer skills: Knowledge of $\text{\LaTeX}$ editing is appreciated.

References

- [1] P.K. Job and M. Srinivasan. Evaluation of reactivity bonus due to $(n, 2n)$ multiplication in Be/BeO-reflected ^{233}U uranyl nitrate solution systems. *Annals of Nuclear Energy*, 9(4):209–213, 1982.
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